

① a) $v(r) = c/r$

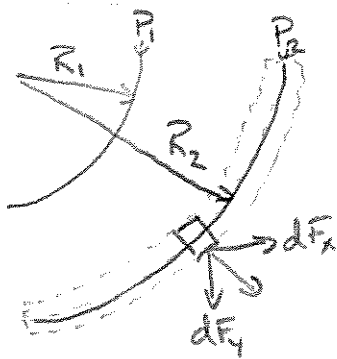
$$\frac{dp}{dr} = \rho \frac{v^2}{r} \quad \text{as derived in lecture}$$

$$\frac{dp}{dr} = \rho \frac{c^2}{r^3} \Rightarrow \int_{P_1}^{P_2} dp = \int_{R_1}^{R_2} \rho \frac{c^2}{r^3} dr$$

$$\Delta p = p_2 - p_1 = \rho c^2 \left(\frac{-1}{R_2^2} + \frac{1}{R_1^2} \right)$$

$$\boxed{\Delta p = \frac{\rho c^2}{R_1^2} \left(1 - R_1^2/R_2^2 \right)}$$

b)



$$dF = \Delta p R_2 d\theta$$

$$dF_x = \Delta p R_2 \sin\theta d\theta$$

$$dF_y = \Delta p R_2 \cos\theta d\theta$$

$$F_x = \Delta p R_2 \int_0^{\pi/2} \sin\theta d\theta = \Delta p R_2$$

$$F_y = \Delta p R_2 \int_0^{\pi/2} \cos\theta d\theta = \Delta p R_2$$

(2)

Continuity $\leftarrow$ Given

$$\begin{aligned} A_1 V_1 &= A_2 V_{2a} \\ A_1 V_1 &= A_2 V_{2b} \end{aligned} \Rightarrow \boxed{V_{2a} = V_{2b}}$$

a) Bernoulli:

$$P_1 + \frac{1}{2} \rho V_1^2 = P_{2a} + \frac{1}{2} \rho V_{2a}^2, \text{ cont: } V_{2a}^2 = \left(\frac{A_1}{A_2}\right)^2 V_1^2$$

$$\boxed{\Delta P_a = P_1 - P_{2a} = \frac{1}{2} \rho V_1^2 \underbrace{\left[\left(\frac{A_1}{A_2}\right)^2 - 1 \right]}_{K_a}}$$

b SFME

$$\rho V_{2b} V_{2b} A_2 - \rho V_1 V_1 A_1 = P_1 A_2 - P_2 A_2$$

$$\Delta P_b = P_1 - P_2 = \frac{2}{A_2} \left(-\frac{1}{2} \rho V_1^2 A_1 + \frac{1}{2} \rho V_{2b}^2 A_2 \right); V_{2b}^2 = \left(\frac{A_1}{A_2}\right)^2 V_1^2$$

$$\boxed{\Delta P_b = \frac{1}{2} \rho V_1^2 \underbrace{\left[2 \left(\frac{A_1}{A_2}\right)^2 - 2 \frac{A_1}{A_2} \right]}_{K_b}}$$

3

a) Continuity: $v \pi \frac{d^2}{4} = v_{top} \pi \frac{D^2}{4} = \left| \frac{dh}{dt} \right| \pi \frac{D^2}{4}$

$$\boxed{\frac{dh}{dt} = - \left(\frac{d}{D} \right)^2 v}$$

Quasisteady if $dh/dt \ll v$, i.e. $d \ll D$

b) Bernoulli:

$$\frac{1}{2} \rho v_{top}^2 + \rho g h = \frac{1}{2} \rho v^2$$

$$v^2 - v_{top}^2 = 2gh$$

$$v_{top}^2 \left[\left(\frac{D}{d} \right)^4 - 1 \right] = 2gh$$

$$\frac{dh}{dt} = - \sqrt{\frac{2gh}{\left(\frac{D}{d} \right)^4 - 1}} = - 2 \sqrt{\frac{g d^4}{2(D^4 - d^4)}} \sqrt{h} = - 2Gh^{1/2}$$

integrating: $h^{-1/2} dh = - 2G dt$

$$\boxed{h^{1/2} = H^{1/2} - Gt} ; t = T_0 \rightarrow h = 0 \Rightarrow \boxed{T_0 = \frac{H^{1/2}}{G}}$$

c) Bernoulli:

$$\frac{1}{2} \rho v_{top}^2 + \rho g (h+L) = \frac{1}{2} \rho v^2$$

same as (b), but with height ' $h+L$ '

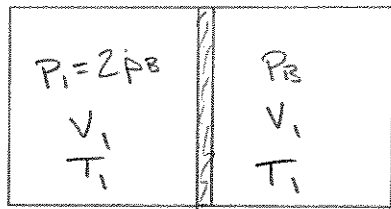
$$(h+L)^{1/2} = (H+L)^{1/2} - Gt$$

$$t = T_0' \rightarrow h = 0$$

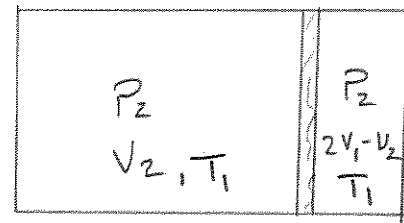
$$T_0' = \frac{(H+L)^{1/2} - L^{1/2}}{G} ; L = H \Rightarrow \boxed{T_0' = \frac{(\sqrt{2}-1)H^{1/2}}{G} \approx 0.41 T}$$

4(a)

Initial



Final



2018 AMB

System A $\textcircled{I} \rightarrow \textcircled{F}$

$$W_A = \int_{V_1}^{V_2} p dV \quad \therefore W_A > 0 \quad \underline{\text{positive}}$$

$>0 \quad >0$

Cons. of energy A

$$Q_A - W_A = \Delta U = C_V \Delta T = 0$$

$$\therefore Q_A = W_A > 0 \quad \underline{Q_A \text{ positive}}$$

System B

$$W_B = \int_{V_1}^{V_2} p dV \quad \therefore W_B < 0 \quad \underline{\text{negative}}$$

$>0 \quad <0$

$$\text{Cons. of Energy } Q_B = W_B \quad \underline{\text{negative}}$$

b) Ideal gas

$$[1] \quad P_{A1} V_{A1} = m_A R_A T_1$$

$$[3] \quad P_{B1} V_1 = m_B R_B T_1 = \frac{1}{2} P_{A1} V_1$$

$$[2] \quad P_{A2} V_{A2} = m_A R_A T_1$$

$$[4] \quad P_{B2} V_{B2} = m_B R_B T_1 = P_{A2} (2V_1 - V_{A2})$$

$$[4] = [3] \quad \frac{1}{2} P_{A1} V_1 = P_{A2} (2V_1 - V_{A2})$$

$$[1] = [2] \quad P_{A1} V_1 = P_{A2} V_{A2} \Rightarrow P_{A2} = P_{A1} V_1 / V_{A2}$$

2 Eqn
2 unknowns
 P_{A2}, V_{A2}

combine to eliminate P_{A2}

$$\frac{1}{2} P_{A1} V_1 = P_{A1} V_1 / V_{A2} (2V_1 - V_{A2})$$

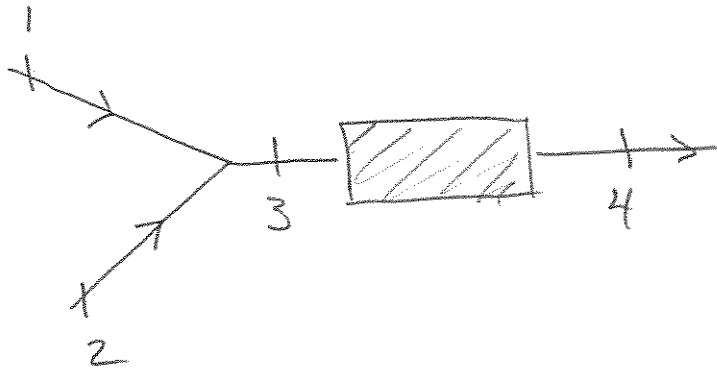
$$\frac{1}{2} V_{A2} = 2V_1 - V_{A2} \Rightarrow \frac{3}{2} V_{A2} = 2V_1 \quad V_{A2} = \frac{4}{3} V_1$$

$$\text{w/ } V_1 = 1 \text{ m}^3$$

$$\boxed{V_{A2} = \frac{4}{3} \text{ m}^3}$$

Note: Applying conservation of energy must include the loss of kinetic energy by a non-conservative force (friction) or the piston will oscillate forever.

P5)



a) $\dot{m}_3 h_3 - \dot{m}_1 h_1 - \dot{m}_2 h_2 = 0$ SFEE

$$\dot{m}_1 c_p (T_3 - T_1) + \dot{m}_2 c_p (T_3 - T_2)$$

$$T_3 = \frac{\dot{m}_1 c_p T_1 + \dot{m}_2 c_p T_2}{(\dot{m}_1 + \dot{m}_2) c_p} = \boxed{1750 \text{ K}}$$

$$\dot{m}_4 h_4 - \dot{m}_3 h_3 = \dot{Q} = \dot{m}_3 (c_p (T_4 - T_3))$$

$$\dot{m}_3 = \dot{m}_4 = 3 \text{ kg/s}$$

$$T_4 = \frac{\dot{Q}}{\dot{m}_3 c_p} + T_3 = \frac{-1,550 \frac{\text{kJ}}{\text{s}}}{4 \frac{\text{kg}}{\text{s}} \cdot 0.35 \frac{\text{kJ}}{\text{kg K}}} + 1750 \text{ K}$$

$$\boxed{T_4 = 642.9 \text{ K}}$$

b) $\frac{dS_{\text{rev}}}{dt} + \sum \dot{m}_{\text{out}} S_{\text{out}} - \sum \dot{m}_{\text{in}} S_{\text{in}} = \int \frac{d\dot{Q}}{T} + \dot{S}_{\text{irr}}$

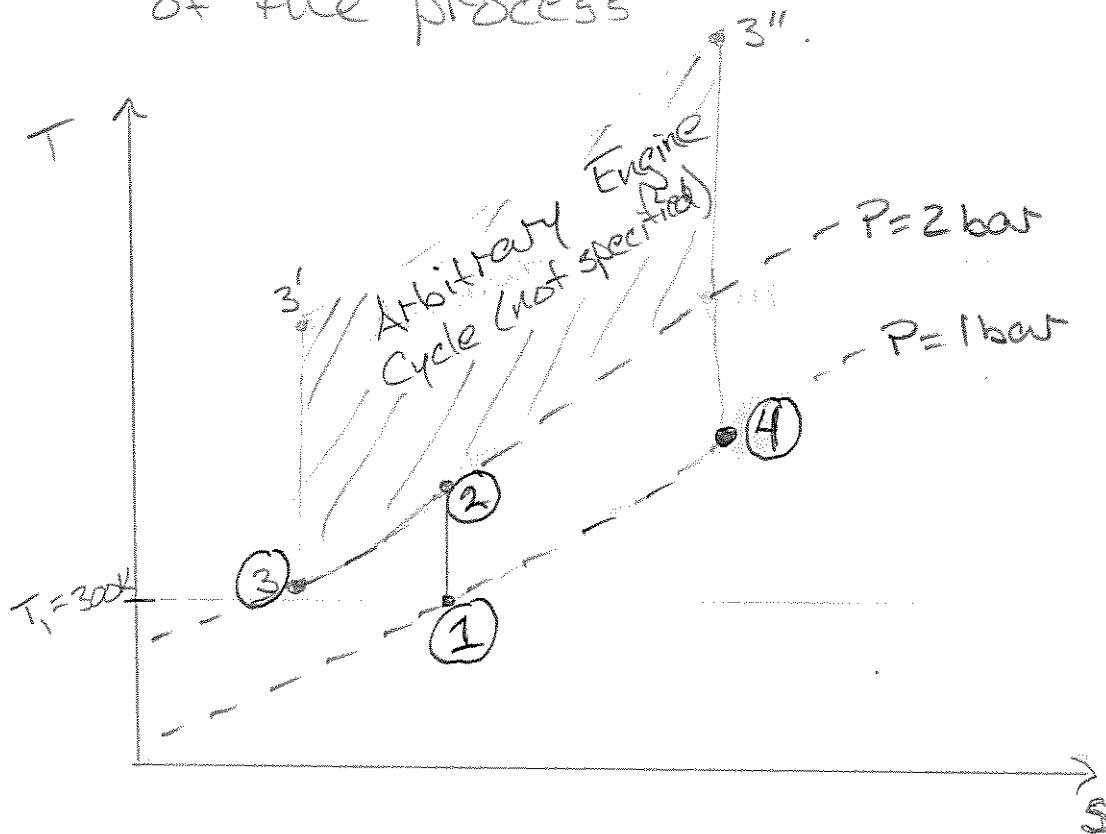
$$dS = c_p \ln(T_2/T_1) - R \ln(P_2/P_1)$$

$$\dot{m}_1 c_p \ln(T_3/T_1) + \dot{m}_2 c_p \ln(T_3/T_2) = \dot{S}_{\text{irr}}$$

$$\dot{S}_{\text{irr}} = 1 \frac{\text{kg}}{\text{s}} 350 \frac{\text{J}}{\text{kg K}} \ln(1750/1000) + 3 \frac{\text{kg}}{\text{s}} 350 \frac{\text{J}}{\text{kg K}} \ln(1750/2000)$$

$$\boxed{\dot{S}_{\text{irr}} = 55.66 \frac{\text{J}}{\text{K s}}}$$

6a) Sketch a T-s diagram of the process



b) SFEE

$$-w_{1-2} = h_2 - h_1 = c_p (T_2 - T_1)$$

Isentropic compression

$$T_2 = T_1 (P_2/P_1)^{\gamma-1/\gamma} = 300 \text{ K} (2)^{0.4/1.4} = \boxed{365.7 \text{ K}}$$

$$\therefore w_{1-2} = -c_p (T_2 - T_1) = -1.005 \frac{\text{kJ}}{\text{kg K}} \cdot 300 \text{ K} (365.7 \text{ K} - 300 \text{ K})$$

$$\boxed{w_{1-2} = -66 \frac{\text{kJ}}{\text{kg}}}$$

$$c) \rho_2 = \frac{P_2}{RT_2} = \frac{2 \cdot 10^5 \text{ Pa}}{287 \frac{\text{J}}{\text{kg K}} \cdot 365.7 \text{ K}} = \boxed{1.906 \frac{\text{kg}}{\text{m}^3}}$$

$$\rho_3 = \frac{P_3}{RT_3} = \frac{P_2}{RT_3} \quad \text{where} \quad q_{ic} = h_3 - h_2 = c_p (T_3 - T_2)$$

$$T_3 = \frac{q_{ic}}{c_p} + T_2 = \frac{-60 \text{ kJ/kg}}{1.005 \frac{\text{kJ}}{\text{kg K}}} + 365.7 \text{ K} = 306 \text{ K}$$

$$\rho_3 = \frac{2 \cdot 10^5 \text{ Pa}}{287 \frac{\text{J}}{\text{kg K}} \cdot 306 \text{ K}} = \boxed{2.277 \frac{\text{kg}}{\text{m}^3}}$$

$$6d) \dot{W} = \dot{Q}_{3-4} - \dot{m} c_p (T_4 - T_3) = \rho \dot{V} (q_{3-4} - c_p (T_4 - T_3))$$

$$\dot{W}_{3-4, No-IC} = \rho_2 \dot{V} (q_{3-4} - c_p (T_4 - T_2)) =$$

$$1.906 \frac{\text{kg}}{\text{m}^3} \cdot 0.1 \frac{\text{m}^3}{\text{s}} \left(10^6 \frac{\text{J}}{\text{kg}} - 1005 \frac{\text{J}}{\text{kgK}} (900\text{K} - 365.7\text{K}) \right)$$

$$\boxed{\dot{W}_{3-4, No-IC} = 88.2 \text{ kW}}$$

$$\dot{W}_{3-4, wIC} = \rho_3 \dot{V} (q_{3-4} - c_p (T_4 - T_3))$$

$$= 2.277 \frac{\text{kg}}{\text{m}^3} \cdot 0.1 \frac{\text{m}^3}{\text{s}} \left(10^6 \frac{\text{J}}{\text{kg}} - 1005 \frac{\text{J}}{\text{kgK}} (900\text{K} - 306\text{K}) \right)$$

$$\boxed{\dot{W}_{3-4, wIC} = 91.8 \text{ kW}}$$

$$e) \eta_{wsc} = \frac{\dot{W}_{34} + \dot{W}_{12}}{\dot{Q}_{34}} = \frac{\cancel{\rho_3 \dot{V}} (q_{34} - c_p (T_4 - T_3)) + \cancel{\rho_3 \dot{V}} w_{12}}{\cancel{\rho_3 \dot{V}} q_{34}}$$

$$\boxed{\eta_{wsc} = 33.7\%}$$

$$\eta_{NoSC} = \frac{q_{34} - c_p (T_4 - T_1)}{q_{34}} = \boxed{39.7\%}$$

(No compressor or intercooler)

f) No intercooler or comp.

$$\dot{W} = \rho_1 \dot{V} (q_{3-4} - c_p (T_4 - T_1)) = \underline{46.1 \text{ kW}}$$

$$\eta = \underline{39.7\%} \quad (e)$$

with compressor and no intercooler

$$\dot{W} = 88.2 \text{ kW} - \rho_2 \dot{V} w_{12} = \underline{75.65 \text{ kW}}$$

$$\eta = \frac{(q_{34} - c_p (T_4 - T_2)) + \dot{w}_2}{q_{34}} = \underline{39.7\%}$$

with comp. and ^{q_{34}} intercooler

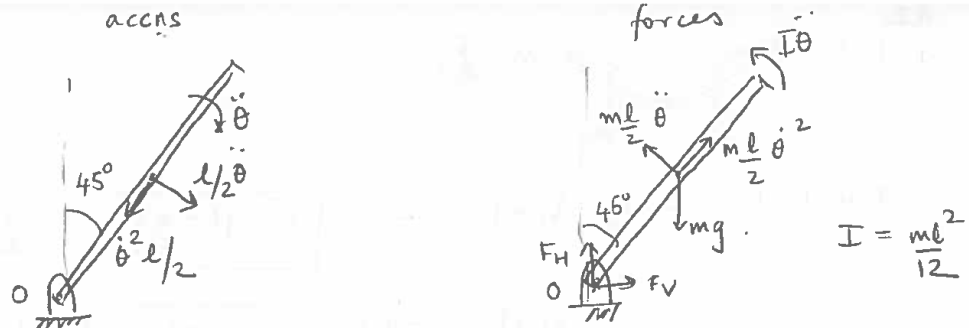
$$\dot{W} = 91.8 \text{ kW} - \rho_3 \dot{V} w_{12} = \underline{76.7 \text{ kW}} \quad \eta = 33.7\%$$

The compressor increases power output with no impact on efficiency due to the isentropic increase in density. The intercooler removes heat and increases work, but decreases η .

1A Mechanical Engineering

Cribs – Section B (AA Seshia and J Biggins)

7 (a)

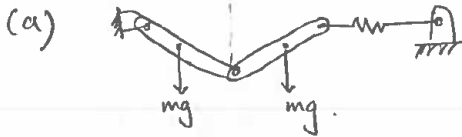


$$\begin{aligned} \Sigma M(O) \Rightarrow \quad & mg \frac{l}{2\sqrt{2}} - m \frac{l^2}{4} \ddot{\theta} - \frac{ml^2}{12} \ddot{\theta} = 0 \\ & \frac{ml^2}{3} \ddot{\theta} = \frac{mg l}{2\sqrt{2}} \\ & \ddot{\theta} = \frac{3g}{2\sqrt{2} l} \end{aligned}$$

(b) Using conservation of energy:

$$\begin{aligned} mg \left(\frac{l}{2} - \frac{l}{2\sqrt{2}} \right) &= \frac{1}{2} \left(\frac{ml^2}{12} \right) \dot{\theta}^2 + \frac{1}{2} m \left(\dot{\theta} \frac{l}{2} \right)^2 \\ mg l \left(\frac{\sqrt{2}-1}{\sqrt{2}} \right) &= \frac{ml^2 \dot{\theta}^2}{3} \\ \therefore \dot{\theta} &= \sqrt{\frac{3g}{l} \left(1 - \frac{1}{\sqrt{2}} \right)} \end{aligned}$$

8



Potential energy $V(x) = -2mg \frac{\sqrt{L^2 - (L-x)^2}}{2} + \frac{1}{2} kx^2$

$$V(x) = -mg \sqrt{\frac{x^2}{4} + xL} + \frac{1}{2} kx^2$$

(b)
$$V'(x) = kx - \frac{mg(L - x/2)}{2\sqrt{xL - x^2/4}}$$

at $\alpha = 45^\circ$, $L - \frac{x}{2} = \frac{1}{\sqrt{2}} L$ or $x = (\sqrt{2} + 2)L$

$$\therefore k \cdot (2 - \sqrt{2})L = \frac{mg L / \sqrt{2}}{2 L / \sqrt{2}}$$

$$\therefore k = \frac{mg}{2(2 - \sqrt{2})L}$$

(c)
$$V''(x) = k + \frac{mg \left(\frac{L-x}{2} \right) \left(L - \frac{x^2}{4} \right)^{-3/2}}{4} + \frac{mg \left(L - \frac{x^2}{4} \right)^{-1/2}}{4} > 0$$

$\therefore$ equilibrium is stable.

9 (a) For a disk of radius r , $I = (1/2)mr^2 = (1/2)\rho t \pi r^4$.

Therefore, for an annulus, $I = (1/2)\rho t \pi (a^4 - b^4) = (1/2)M(a^2 + b^2)$.

The weights in the arms are a total distance $2(a+b)$ from the axis

So, by the parallel axis theorem, each contributes $I = (1/2)M(a^2 + b^2) + M(2a + 2b)^2$ to the total moment of inertia. The total moment of inertia is $\frac{1}{2}M(4(a^2 + b^2)) + 3(2a + 2b)^2 = 2M(7a^2 + 7b^2 + 12ab)$.

(b) The weight has $r = 2(a+b)$ is constant, and $\dot{\theta} = ct$

From the data book, radial acceleration in circular motion is $-r\dot{\theta}^2 = -2(a+b)c^2t^2$

From the data book, angular acceleration in circular motion is $r\ddot{\theta} = 2(a+b)c$

So using $\mathbf{F} = m\mathbf{a}$, the total force is $\mathbf{F} = 2m(a+b)c(\mathbf{e}_\theta - ct^2\mathbf{e}_r)$.

10 (a) Using polar coordinates and databook -
 $\vec{a}_{\text{radial}} = -\frac{GM}{r^2} \hat{e}_r = (\ddot{r} - r\dot{\theta}^2) \hat{e}_r$

$$\vec{a}_{\text{tangential}} = 0 = (r\ddot{\theta} + 2\dot{r}\dot{\theta}) \hat{e}_{\theta}$$

$$\begin{aligned} \frac{d}{dt}(r^2\dot{\theta}) &= 2r\dot{r}\dot{\theta} + r^2\ddot{\theta} \\ &= r(2\dot{r}\dot{\theta} + r\ddot{\theta}) = 0 \end{aligned}$$

$\therefore mr^2\dot{\theta}$ is a constant

$\therefore$ Angular momentum is a constant.

(b) $\frac{d^2}{d\theta^2}\left(\frac{1}{r}\right) + \left(\frac{1}{r}\right) = \frac{gR_E^2}{r_0^2 v_0^2}$

$\therefore$ if $\frac{1}{r} = u$ then

$$\frac{d^2 u}{d\theta^2} + u = \frac{gR_E^2}{r_0^2 v_0^2} \quad \text{is a 2nd order ODE in } u$$

$$u = A \cos \theta + B \sin \theta + \frac{gR_E^2}{r_0^2 v_0^2}$$

at $\theta = 0$, $u = 1/r_0$

$$\therefore A = \frac{1}{r_0} - \left(\frac{gR_E^2}{r_0^2 v_0^2} \right)$$

at $\theta = 0$, $\frac{du}{d\theta} = B = \frac{1}{\dot{\theta}} \frac{du}{dt} = \frac{1}{\dot{\theta}} \frac{d}{dt} \left(\frac{1}{r} \right)$

$$= -\frac{1}{r^2 \dot{\theta}} (\dot{r}) = 0 \quad (\text{as } \dot{r} = 0 \text{ at } \theta = 0)$$

$$\therefore \frac{1}{r} = \frac{gR_E^2}{r_0^2 v_0^2} (1 + e \cos \theta)$$

where $e = \left[\frac{1}{r_0} - \frac{gR_E^2}{r_0^2 v_0^2} \right] / \frac{gR_E^2}{r_0^2 v_0^2}$

By comparison to the Mechanics Databook.

(c) From the databook

$$r_A = \left(\frac{1+e}{1-e} \right) r_P = \frac{1}{r_0} \times r_0 = \frac{r_0^2 v_0^2}{2gR_E^2 - v_0^2 r_0}$$

$$v_A = \frac{v_0 r_0}{r_A} = \frac{2gR_E^2 - v_0^2 r_0}{v_0 r_0}$$

For circular orbit: $\frac{v_c^2}{r_A} = \frac{GM}{r_A^2} = \frac{gR_E^2}{r_A^2}$

$$\therefore v_c = \sqrt{\frac{gR_E^2}{r_A}} = v_1$$

$$I = m(v_1 - v_A)$$

$$= m \left[\frac{\sqrt{gR_E^2(2gR_E^2 - v_0^2 r_0)} + v_0^2 r_0 - 2gR_E^2}{v_0 v_0} \right]$$

(d) Each of the masses has value $m/2$. Assuming breakup is instantaneous.

$$m v_1 = \frac{m}{2} v_1 + \frac{m}{2} v_2$$

$$\therefore v_2 = -v_1$$

$$\therefore \text{Energy released} = -\frac{1}{2} m v_1^2 + \frac{1}{2} \left(\frac{m}{2} v_1 \right)^2 + \frac{1}{2} \left(\frac{m}{2} v_1 \right)^2$$

$$= \frac{1}{2} m v_1^2 \left(-1 + \frac{1}{2} + \frac{1}{2} \right)$$

$$= 2 m v_1^2$$

$$= \frac{2 m g R_E^2}{r_A} = \frac{2 m g R_E^2 (2gR_E^2 - v_0^2 r_0)}{r_0^2 v_0^2}$$

(a) The length of rope is $l = l_0 + vt - x$ so the spring force is $F_{\text{spring}} = k(vt - x)$ and the dashpot force is $F_{\text{dashpot}} = \lambda \dot{l} = \lambda(v - \dot{x})$.

Applying $ma = F$ to the car, we then get $m\ddot{x} = k(vt - x) + \lambda(v - \dot{x})$.

(b) We substitute $y = x - vt$, to turn the equation into the standard form:

$$m\ddot{y} = -ky - \lambda\dot{y}$$

From the databook, the solution for critical damping with zero initial displacement is

$$y = Ate^{-\sqrt{\frac{k}{m}}t}.$$

At $t = 0$ we have $y = 0$ and $\dot{y} = -v$, so we need $A = -v$. Substituting back for x , the final answer is

$$x = vt \left(1 - e^{-\sqrt{\frac{k}{m}}t} \right).$$

(Many candidates successfully solved this problem using the hint as the complementary function and trial particular integral of the form $x_{PI} = A + Bt$)

12 (a) 3 coordinates (aka degrees of freedom) so expect 3 normal modes.
An up and down mode, a side-to-side mode in x and a side-to-side mode in y .

(b) For the “up-and-down” mode, the equation of motion is

$$m\ddot{z} = k\left(\frac{4}{3}l - z - l\right) - k\left(\frac{4}{3}l + z - l\right) = -2kz.$$

So the mode is simply $(x, y, z) = \left(0, 0, z_0 \cos\left(\sqrt{\frac{2k}{m}}t + \phi\right)\right)$ i.e. it has $\omega_1 = \sqrt{\frac{2k}{m}}$.

(c) When the mass is at $(x, y, z) = (x, 0, 0)$, the spring force is unchanged, $F = kl/3$.
However, this force points along the spring. Since there are two springs the z components of the forces cancel and the x components add.

If θ is the angle the spring makes with the z direction, the total force in the x direction is

$$F = -2\frac{kl}{3} \sin \theta$$

which, using the small angle approximation $\sin(\theta) \approx \tan(\theta)$ is

$$F \approx -2\frac{kl}{3} \frac{x}{4l/3} \approx -\frac{k}{2}x.$$

The equation of motion is thus $m\ddot{x} = -\frac{k}{2}x$.

So the second mode is $(x, y, z) = (x_0 \cos(\sqrt{k/(2m)}t), 0, 0)$, which does indeed have $\omega_2 = \sqrt{k/(2m)} = \omega_1/2$.

(d) The vertical equation of motion is now $m\ddot{z} = -2kz + f_0 \cos\left(\frac{3}{4}\omega_1 t\right)$.

In standard form, this is $\frac{m}{2k}\ddot{z} + z = \frac{f_0}{2k} \cos\left(\frac{3}{4}\omega_1 t\right)$,

so, in databook form, we identify $y = z$, $\omega_n = \sqrt{2k/m}$, $\xi = 0$ and $X = \frac{f_0}{2k}$

Reading the data-book at $\omega = 0.75\omega_n$, we see $|Y|/|X| \approx 2.3$ and $\phi = 0$, so the final motion is $z = 2.3\frac{f_0}{2k} \cos\left(\frac{3}{4}\omega_1 t\right)$

(e) The new equation of motion is $m\ddot{z} = -2kz - \lambda\dot{z} + f_0 \cos\left(\frac{3}{4}\omega_1 t\right)$

Putting this in standard form, $\frac{m}{2k}\ddot{z} + \frac{\lambda}{2k}z = \frac{f_0}{2k} \cos\left(\frac{3}{4}\omega_1 t\right)$

so, in databook form, we identify $y = z$, $\omega_n = \sqrt{2k/m}$, $\xi = \lambda/(2\sqrt{2km})$ and $X = \frac{f_0}{2k}$

We want $|Y|/|X| \approx 2.3/2 \approx 1.15$, which, from the databook graph, requires $\xi \approx 0.5$ so, we must choose $\lambda \approx \sqrt{2km}$.

END OF PAPER